

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 5: L6

Question 1 (Lecture 6)

Which of the following functions are convex? Select all that apply.

- (a) $f(x) = 5 - (x - 2)^2, x \in \mathcal{R}$
- (b) $f(x) = 7 - 3x, x \in \mathcal{R}$
- (c) $f(x) = \log(x), x \in (0, +\infty)$
- (d) $f(x) = e^x, x \in \mathcal{R}$
- (e) $f(x) = |2x + 2|, x \in \mathcal{R}$

Question 2 (Lecture 6)

True or false: For a convex optimization problem with a differentiable objective function, gradient descent always converges to the global minimum if the learning rate is chosen appropriately.

- (a) True
- (b) False

Question 3 (Lecture 6)

Which of the following statements about gradient descent are correct? Select all that apply.

- (a) Gradient descent moves in the direction of steepest descent, which is the negative gradient of the function.
- (b) For convex functions with a properly chosen learning rate, gradient descent will converge to the global minimum.
- (c) If the learning rate is too large, gradient descent may overshoot and fail to converge.
- (d) Gradient descent is guaranteed to find the global minimum even for non-convex functions.
- (e) Gradient descent only works for functions that are differentiable and strictly convex.

Question 4 (Lecture 6)

Suppose we are minimizing $f(x) = x^6$ with respect to x . We initialize $x = 1$, and perform gradient descent with learning rate $\eta = 0.01$.

- (i) Calculate the updated value of x after the second iteration. Round your answer to 2

decimal places.

- (ii) Explain why the updates in gradient descent become very small as x approaches 0. How does the shape of the function $f(x) = x^6$ influence the speed of convergence compared to a function like $f(x) = x^2$?

Question 5 (Lecture 6)

Suppose we have a function that depends on a vector of numbers $\mathbf{w}$:

$$C(\mathbf{w}) = \sum_{i=1}^m (\mathbf{w}^T \mathbf{x}_i - y_i)^2$$

where:

$\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ are given vectors,

$y_1, y_2, \dots, y_m$ are given numbers (i.e., scalars).

We want to minimize $C(\mathbf{w})$ using gradient descent.

- (i) Check if the function $C(\mathbf{w})$ is convex. Explain your reasoning.
- (ii) Compute the gradient of $C(\mathbf{w})$ with respect to $\mathbf{w}$ using $\sum$ notation.
- (iii) Write the gradient descent update rule for $\mathbf{w}$ with learning rate η .
- (iv) If $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\mathbf{x}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$, $\eta = 0.01$, we initialize $\mathbf{w} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, calculate the updated value of $\mathbf{w}$ after the first iteration.
- (v) (*Optional*): Following the setup in (iv), solve the convex optimization problem using a solver (e.g., CVXPY).